

Prediction of Coronary Heart Disease - Logistic Analysis

Cardiovascular Disease(CVD) is a leading cause of morbidity and mortality worldwide. It's influences are multifaceted, including demographic, lifestyle, and medical factors. Given its widespread impact, CVD remains a critical focus for public health initiatives globally. One of the most prevalent types of CVD is Coronary Heart Disease(CHD), which occurs when the coronary arteries, supplying blood to the heart, become narrow or blocked, usually due to build up of fatty deposits. This restricts blood flow to the heart, leading to symptoms like chest pain and shortness of breath, or more severe issues like heart attacks and strokes. CHD is responsible for a significant portion of CVD-related deaths and is the center of many studies.

In the Literature Review completed previously, it was found that several factors can contribute to the development of CHD. Smoking, especially long-term tobacco use, is strongly linked to an increased risk of the condition. It is also the leading cause of heart disease among individuals with diabetes. Both high and low blood pressure have been associated with an elevated risk. As well as the rise in obesity has been paralleled to an increase in cases. Additionally, Age and gender also play significant roles in the prevalence of the disease. This report will delve further into these elements and assess their relative impact on the development of CHD.

In this analysis, Logistic Regression will be used. The predictor variables of interest are age, gender, smoking, obesity/body mass index(BMI), diabetes, and blood pressure. Data from the Framingham Heart Study will be utilized. The Framingham Heart Study began in 1948 with aims to examine the incidence and trends for CHD and still continues to this day. Subjects in the study where chosen at random, any adult over the age of 30 in Framingham, MA could participate in the experiment, as long as they did not have a history of heart disease. Our data is comprised of over 4,000 records of patient data, with 15 variables that represent widely recognized contributors to CHD, including our variables of interest. The response variable is binary, indicating whether a person is at risk of developing CHD within 10 years. The objective of this analysis is to uncover meaningful patterns, build a regression model to highlight which factors are the most influential and determine if we can predict the risk of CHD with strong accuracy.

Beginning the assessment by getting familiar with the data. The dataset has been cleaned and should not have any missing values. Importing and checking the upload below.

```
#Importing the data for Framingham Heart Study
options(repos = c(CRAN = "https://cloud.r-project.org/"))
FH_Heart <- read.csv("C:\\Users\\knigh\\Downloads\\framingham.processed.csv")
head(FH_Heart)
```

```
##   gender age education currentSmoker cigsPerDay BPMeds prevalentStroke
## 1     1  39         1             0           0       0              0
## 2     0  46         0             0           0       0              0
## 3     0  62         1             1          20       1              0
## 4     0  46         0             1          15       0              0
## 5     1  47         0             1          20       0              0
## 6     0  43         0             0           0       0              0
##   prevalentHyp diabetes totChol sysBP diaBP   BMI heartRate glucose TenYearCHD
## 1             0         0     195  106   70 26.97      80      77        0
## 2             0         0     250  121   81 28.73      95      76        0
## 3             1         1     358  215  110 37.62     110     368        1
## 4             0         1     233  106   60 20.84      75     348        0
## 5             1         1     296  141   93 28.50      68     332        1
## 6             1         0     228  180  110 30.30      77      99        0
```

```
str(FH_Heart) #checking the structure of the dataset
```

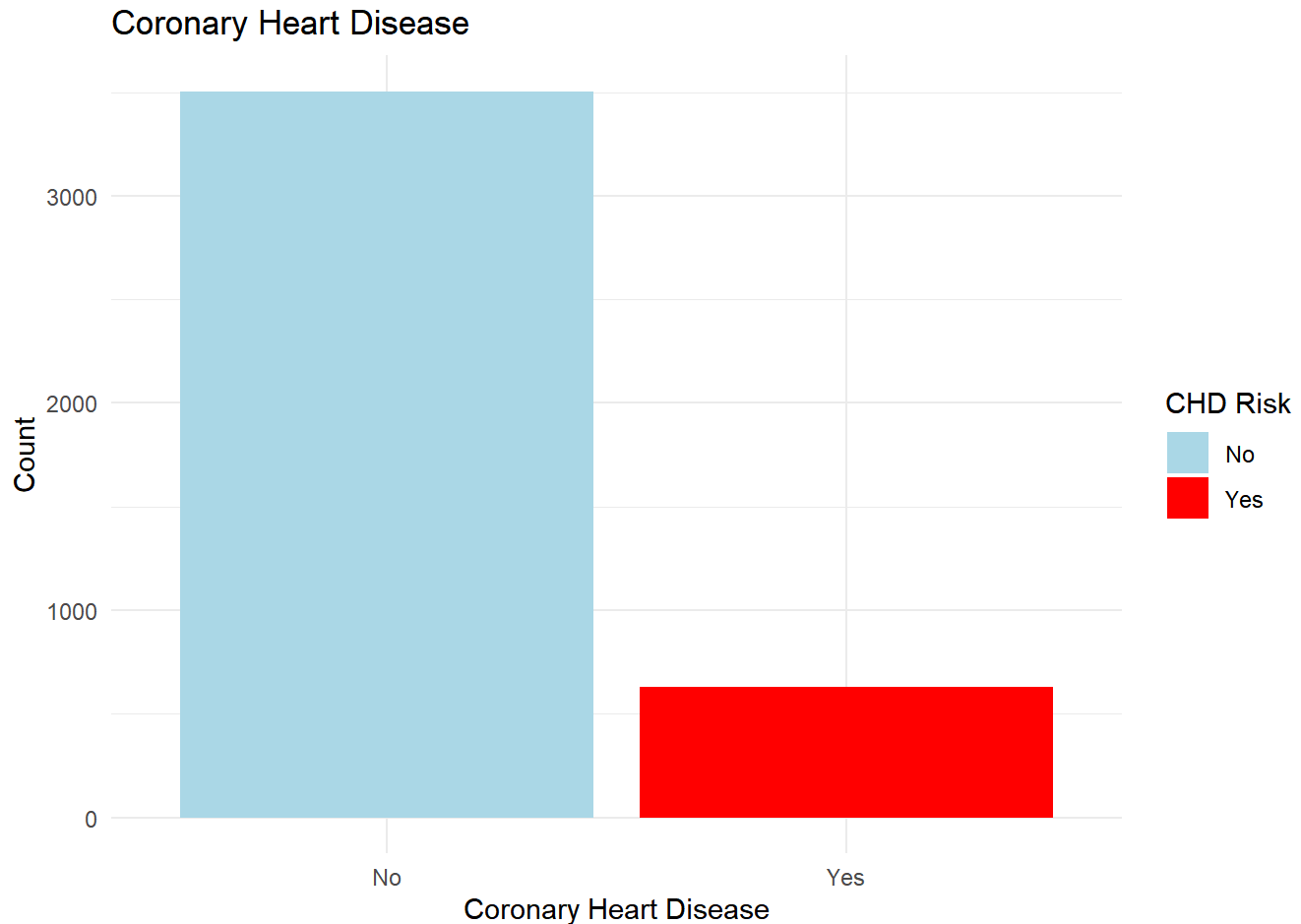
```
## 'data.frame':   4133 obs. of  16 variables:
## $ gender      : int  1 0 0 0 1 0 0 1 1 0 ...
## $ age         : int  39 46 62 46 47 43 63 55 52 47 ...
## $ education   : int  1 0 1 0 0 0 0 0 0 0 ...
## $ currentSmoker : int  0 0 1 1 1 0 0 1 0 1 ...
## $ cigsPerDay   : int  0 0 20 15 20 0 0 40 0 1 ...
## $ BPMeds       : int  0 0 1 0 0 0 0 0 0 0 ...
## $ prevalentStroke: int  0 0 0 0 0 0 0 0 0 0 ...
## $ prevalentHyp : int  0 0 1 0 1 1 0 0 1 1 ...
## $ diabetes     : int  0 0 1 1 1 0 0 1 0 1 ...
## $ totChol      : int  195 250 358 233 296 228 205 205 260 160 ...
## $ sysBP        : num  106 121 215 106 141 ...
## $ diaBP        : num  70 81 110 60 93 110 71 76 89 109 ...
## $ BMI          : num  27 28.7 37.6 20.8 28.5 ...
## $ heartRate    : int  80 95 110 75 68 77 60 90 76 82 ...
## $ glucose      : int  77 76 368 348 332 99 85 325 79 320 ...
## $ TenYearCHD   : int  0 0 1 0 1 0 1 1 0 1 ...
```

```
colSums(is.na(FH_Heart)) #confirming the dataset does not have any NA's
```

```
##           gender           age           education   currentSmoker   cigsPerDay
##           0             0             0             0             0
##   BPMeds prevalentStroke   prevalentHyp         diabetes         totChol
##           0             0             0             0             0
##   sysBP      diaBP           BMI      heartRate         glucose
##           0             0             0             0             0
##   TenYearCHD
##           0
```

As the dataset has over 4,000 records, it is valuable to visualize the target variable.

```
## package 'car' successfully unpacked and MD5 sums checked
##
## The downloaded binary packages are in
## C:\Users\knigh\AppData\Local\Temp\RtmpcRx9Ya\downloaded_packages
```

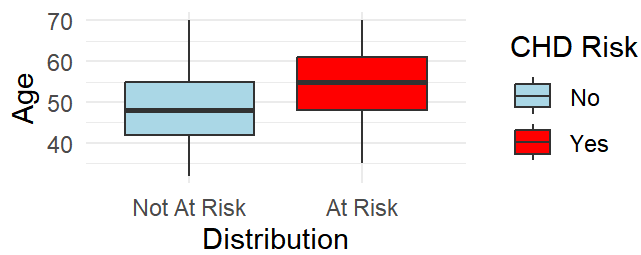


The graph illustrates the proportion of participants in the study classified as at risk for CHD within the next 10-years compared to those not at risk. Out of 4,133 participants, 18% are predicted to be at risk.

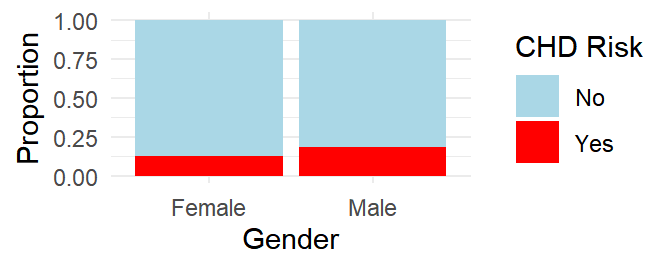
Exploring the key factors contributing to this risk, the graphs below examine how CHD risk correlates with our selected variables, providing meaningful insights into the underlying causes of the disease. The dataset has 10 variables that align with our areas of interest.

```
## package 'cowplot' successfully unpacked and MD5 sums checked
##
## The downloaded binary packages are in
## C:\Users\knigh\AppData\Local\Temp\RtmpcRx9Ya\downloaded_packages
```

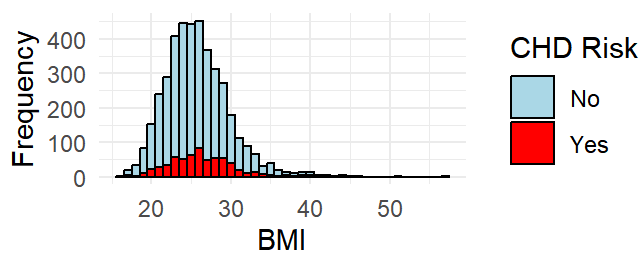
Age Distribution by CHD



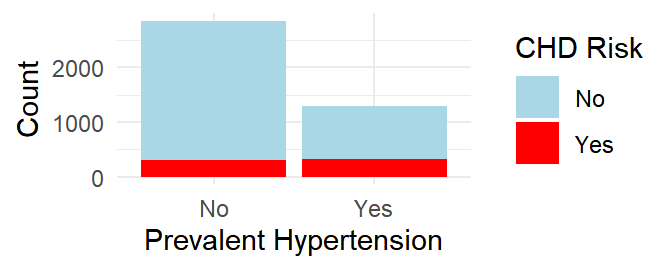
CHD by Gender



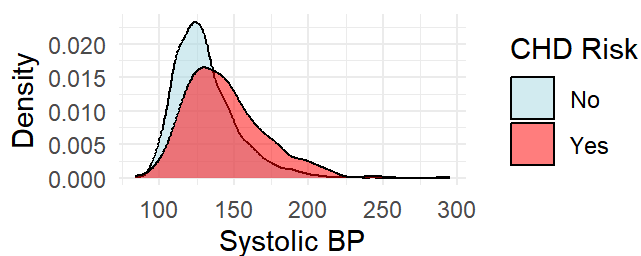
Histogram of BMI



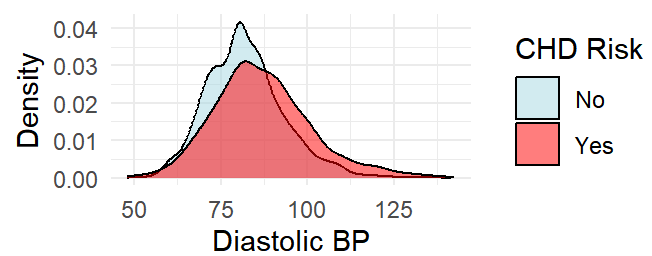
CHD by Hypertension



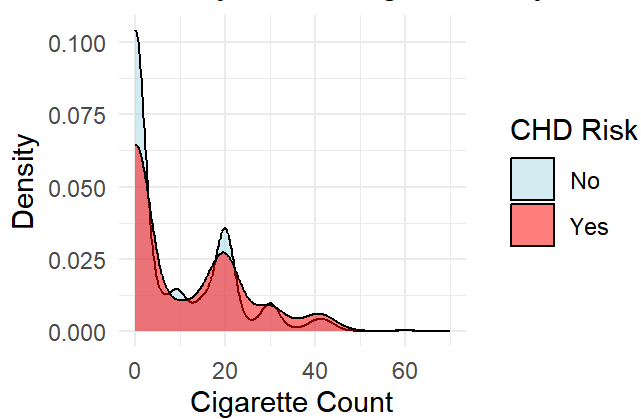
Density Plot of Systolic BP by CHD Risk



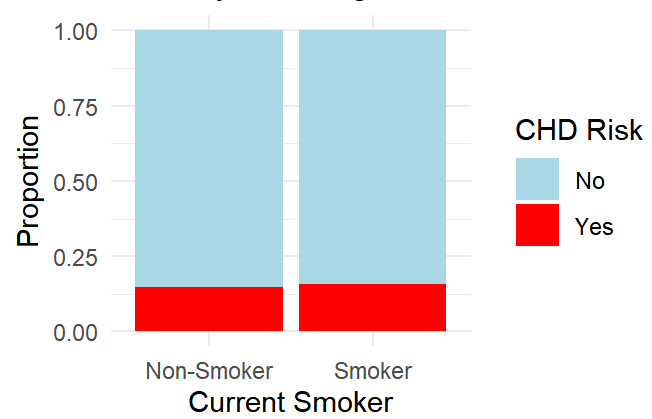
Density Plot of Diastolic BP by CHD Risk



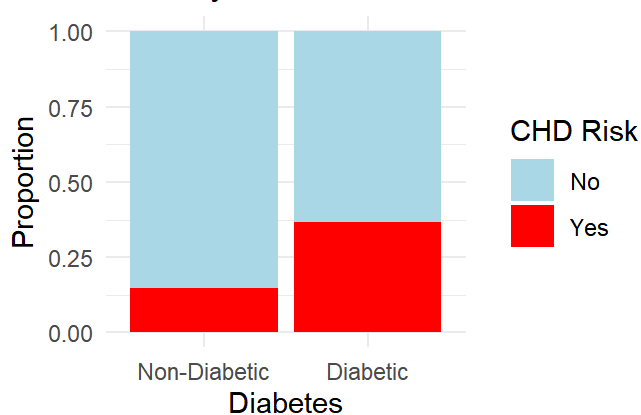
Density Plot of Cigarettes by CHD Risk



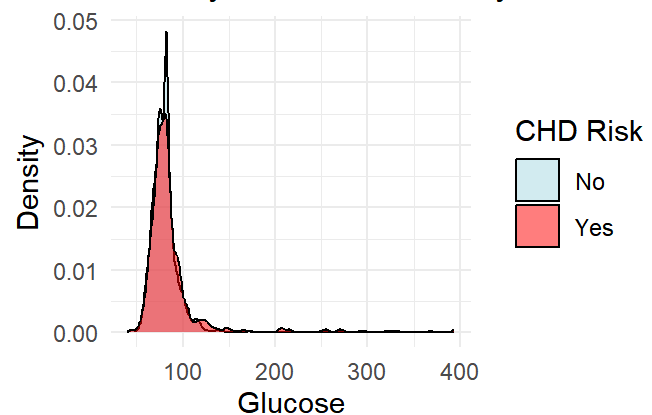
CHD by smoking



CHD by Diabetes

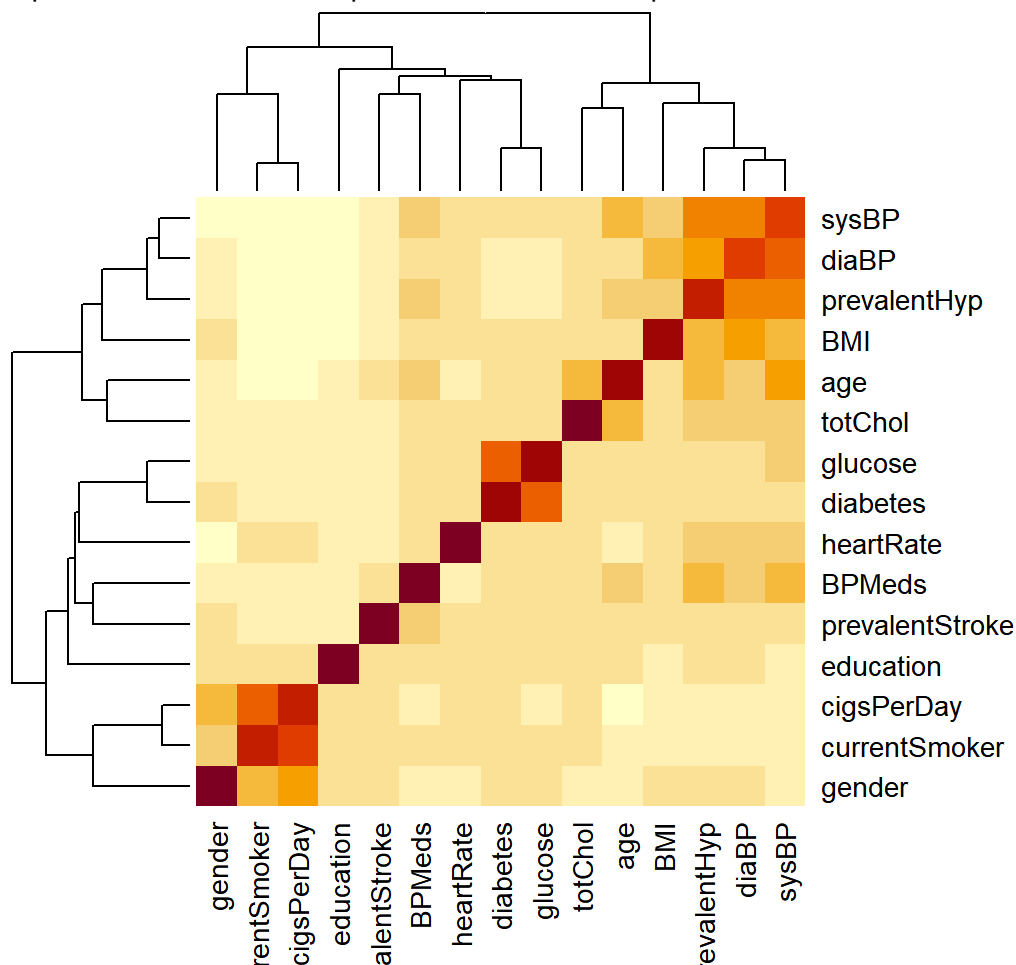


Density Plot of Glucose by CHD Risk



The findings reveal several key insights into the factors influencing CHD risk. Age is a significant factor, with older individuals of 50 and up being more at risk. Men exhibit a higher proportion of CHD risk compared to women. BMI, which follows a near-normal distribution centered around 25 (overweight by CDC standards), also correlates with higher incidence. Participants with hypertension show a noticeably greater chance of developing CHD compared to those without. Elevated systolic (average 143.8 mmHg) and diastolic blood pressures (average 87 mmHg) are associated with increased vulnerability, though diastolic values display more variability. Those smoking over 20 cigarettes daily have higher risk concentrations, and current smokers show greater prevalence of CHD. Diabetics exhibit a strikingly higher CHD risk than non-diabetics, while glucose levels do not significantly differentiate between at-risk and not-at-risk groups due to substantial overlap in the density plot. Seeing these predictor variables interact with our response variable helps in determining which ones to use for our regression model.

Calculating variable correlations can also provide valuable insights into the relationships and interactions within our data. The heatmap below shows how each predictor variable corresponds to one another.



The darker squares on the graph represent areas where correlation between variables are high. Almost half of our variables appear to have darker squares. The strength of the correlations can be calculated. A score of .5 or higher indicates a close relationship. It is valuable to know the strength of our variables relationships, because if predictor variables are highly correlated, it could lead to possible multicollinearity in our model. Below is a list of calculated correlations over the threshold of .5 strength.

##	Var1	Var2	Freq
## 53	cigsPerDay	currentSmoker	0.7717391
## 68	currentSmoker	cigsPerDay	0.7717391
## 123	sysBP	prevalentHyp	0.6974322
## 124	diaBP	prevalentHyp	0.6176692
## 143	glucose	diabetes	0.6043573
## 168	prevalentHyp	sysBP	0.6974322
## 172	diaBP	sysBP	0.7846914
## 184	prevalentHyp	diaBP	0.6176692
## 187	sysBP	diaBP	0.7846914
## 233	diabetes	glucose	0.6043573

There are 5 interactions that exhibit moderate to high correlations. Smoking and cigarettes per day are strongly correlated with a coefficient of 0.7717. Hypertension is strongly tied to systolic blood pressure (0.6974) and moderately to diastolic blood pressure (0.6176). Systolic and diastolic pressures themselves also share a strong relationship (0.7847). Lastly, diabetes and glucose show a moderate correlation of 0.6044. These relationships highlight potential redundancies, as high correlations suggest some variables may carry overlapping information. This can be addressed by selecting fewer predictors, simplifying the model and improving interpretability without losing key insights.

For variable selection, predictors include age, gender, BMI, current smoker status, systolic and diastolic blood pressures, and diabetes. Current smoker was chosen over cigarettes per day to focus on whether smoking itself influences CHD. Similarly, prevalent hypertension was removed as it assumes high blood pressure, whereas both high and low pressures are relevant risks. Despite their high correlation, systolic and diastolic pressures were included as both are important halves of blood pressure readings. The model will be checked for multicollinearity. Additionally, diabetes was selected over glucose, while glucose can signify diabetes, a direct diagnosis better aligns with our areas of interest.

This variable selection prioritizes interpretability and relevance. Moving forward, before fitting the model, the dataset will be divided into training and testing subsets to later evaluate its performance. A 75%-25% split will be applied, with 75% of the data (3,099 rows) allocated to training and 25% (1,034 rows) for testing. The training subset, consisting of randomized rows, will be used to develop the model, while the testing subset will serve as unseen new data to analyze the model's predictive accuracy. This approach will assist with assessing the model's generalizability and reliability in predicting CHD outcomes.

```
set.seed(1)
train <- sample(1:nrow(FH_Heart), size = floor(nrow(FH_Heart)* .75))
test <- FH_Heart[-train,]
train <- FH_Heart[train,]
nrow(train) #3099 rows (75%)
```

```
## [1] 3099
```

```
nrow(test) #1034 rows (25%)
```

```
## [1] 1034
```

Now that the training and testing subsets have been created, the model will be fit below.

```
Model <- glm(TenYearCHD~gender+age+currentSmoker+diabetes+sysBP+diaBP+BMI, data=train, family=binomial)
summary(Model)
```

```
##
## Call:
## glm(formula = TenYearCHD ~ gender + age + currentSmoker + diabetes +
##      sysBP + diaBP + BMI, family = binomial, data = train)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -7.803895   0.552979 -14.112  < 2e-16 ***
## gender         0.455860   0.110180   4.137 3.51e-05 ***
## age           0.060798   0.007051   8.623  < 2e-16 ***
## currentSmoker  0.443689   0.113056   3.925 8.69e-05 ***
## diabetes      0.901909   0.258601   3.488 0.000487 ***
## sysBP         0.016407   0.003838   4.275 1.91e-05 ***
## diaBP         0.002125   0.006929   0.307 0.759100
## BMI           0.002528   0.013535   0.187 0.851835
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 2630.9  on 3098  degrees of freedom
## Residual deviance: 2383.9  on 3091  degrees of freedom
## AIC: 2399.9
##
## Number of Fisher Scoring iterations: 5
```

At first glance, the model summary reveals that most variables are significant, with low standard errors and p-values. Standard errors smaller than the estimate coefficients and p-values under .05 indicate they are likely good predictors. However, there are two variables that do not meet this criteria, diaBP and BMI. Both have standard errors higher than the coefficient estimates and elevated p-values of .76 and .85. Given the high correlation between diastolic and systolic blood pressure, multicollinearity may be a concern. With multicollinearity inflating variance, making variable estimates unstable, we can detect this possible issue using the Variance Inflation Factor (VIF):

```
vif(Model)
```

```
##      gender      age currentSmoker    diabetes    sysBP
##  1.099114  1.234033  1.151828  1.024943  3.148913
##    diaBP      BMI
##  2.883030  1.175824
```

A VIF score of 5 or higher typically indicates multicollinearity has been detected. In this case, all VIF scores are below this threshold, suggesting that despite the high correlation, diastolic pressure does not overlap sufficiently with other variables to cause instability within the overall model. Low p-values and VIF score together suggests that BMI and DiaBP may not contribute significantly to explaining CHD Risk.

BMI, as a general measure of body weight relative to height, becomes significant for CHD primarily at higher levels, such as overweight(25kg/m^2+) and obese(30kg/m^2+) categories. In our data, the average BMI recorded was 25.78kg/m^2 , borderline overweight. This may explain its weak predictive value, as there may be insufficient representation of participants with high BMIs. Similarly, while diastolic pressure reflects the heart's resting state, systolic pressure measures while the heart is in motion during heartbeats. Systolic is more often examined when diagnosing heart disease compared to diastolic. Although extreme diastolic pressures can affect heart health on its own, its role in developing CHD tends to be very small and unlikely.

Given these considerations, it may be best to remove these variables. Akaike Information Criterion (AIC) can help evaluate this. AIC measures a model's overall quality of fit and complexity, with lower values indicating a better model. Comparing the base model to versions excluding DiaBP and BMI could highlight if the model improves.

```
AIC(Model) #Base model
```

```
## [1] 2399.922
```

```
AIC(glm(TenYearCHD~gender+age+currentSmoker+diabetes+sysBP+BMI, data=train, family=binomial)) #D  
iaBP removed
```

```
## [1] 2398.016
```

```
AIC(glm(TenYearCHD~gender+age+currentSmoker+diabetes+sysBP, data=train, family=binomial)) #BMI r  
emoved
```

```
## [1] 2396.074
```

The model's AIC decreases after removing both DiaBP and BMI, confirming an improved fit. The refined model, with the variables age, gender, smoking status, and systolic pressure, appears to be more effective for analyzing CHD risk. Now to further analyze the new simpler model.

```
Model <- glm(TenYearCHD~gender+age+currentSmoker+diabetes+sysBP, data=train, family=binomial)  
summary(Model)
```

```
##
## Call:
## glm(formula = TenYearCHD ~ gender + age + currentSmoker + diabetes +
##       sysBP, family = binomial, data = train)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -7.677276   0.436315 -17.596  < 2e-16 ***
## gender         0.462482   0.108829   4.250 2.14e-05 ***
## age           0.060249   0.006863   8.779  < 2e-16 ***
## currentSmoker  0.437920   0.111564   3.925 8.66e-05 ***
## diabetes      0.903194   0.256979   3.515  0.00044 ***
## sysBP         0.017488   0.002313   7.559 4.06e-14 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 2630.9  on 3098  degrees of freedom
## Residual deviance: 2384.1  on 3093  degrees of freedom
## AIC: 2396.1
##
## Number of Fisher Scoring iterations: 5
```

In the new model, all predictor variables look to be statistically significant, with low standard errors and p-values well below 0.05. The positive coefficient estimates indicate a positive correlation between the predictors and the response variable. This suggests that as the magnitude of these risk factors increase, so does the likelihood of CHD, aligning with expectations for disease related risk factors. This is further highlighted in our response variable's estimate of -7.67, because the intercept reflects the baseline chance of CHD when all predictors are set to 0, the correlation is showing as strongly negative. Meaning without the presence of these risk factors, the probability of CHD is extremely low.

The coefficient estimates provide additional insights when interpreted as odds ratios, quantifying how much more likely CHD is to occur with changes in each predictor. Odds ratios are calculated by exponentiating the coefficient estimates.

```
exp(coef(Model))
```

```
##      (Intercept)          gender          age currentSmoker          diabetes
## 0.0004632351  1.5880099216  1.0621013582  1.5494816235  2.4674726148
##           sysBP
## 1.0176413592
```

To interpret the odds ratios, the binary variables compare the odds for one group versus the other. Gender (male=1 and female = 0) compares men versus women. The calculated value indicates that the odds for men to have CHD risk is 1.5888 times higher or 58.8% more than women. Current smokers have 54.95% higher odds of CHD risk compared to non-smokers, and individuals with diabetes have a 147% increase in odds. For continuous variables, the odds ratio shows how much the odds of CHD change per unit increase in the variable. Systolic blood pressure (sysBP) has an odds ratio of 1.0176, indicating a 1.76% increase in CHD odds for each 1 mmHg rise, while age

increases 6.2% in odds per additional year. These findings provide a good baseline for understanding how each predictor influences CHD risk, with further analysis needed to validate and refine these interpretations as the model is evaluated more thoroughly.

Other methods for evaluating the model include AIC, R^2 , and residuals. The AIC value of 2396, which was calculated earlier, is lower than that of the initial model with DiaBP and BMI, indicating that the current model's predictors offer a better fit for explaining the likelihood of CHD. R^2 and adjusted R^2 can measure how much variability in the response variable is explained by the predictors. While R^2 doesn't apply directly in logistic regression, there is a pseudo R^2 called McFadden's R^2 that compares the likelihood of the model to a null model, one with no predictors. In linear regression, a value of 0.7 or higher is typically considered acceptable, but for logistic models, it is lowered where McFadden's R^2 is considered good above 0.2.

```
#McFadden R2
logLik.model <- logLik(Model) # Log-likelihood of the current model
logLik.null <- logLik(glm(TenYearCHD ~ 1, family = binomial, data = train)) # Log-likelihood of
null model
pseudo.R2 <- 1 - (logLik.model / logLik.null)
print(pseudo.R2)
```

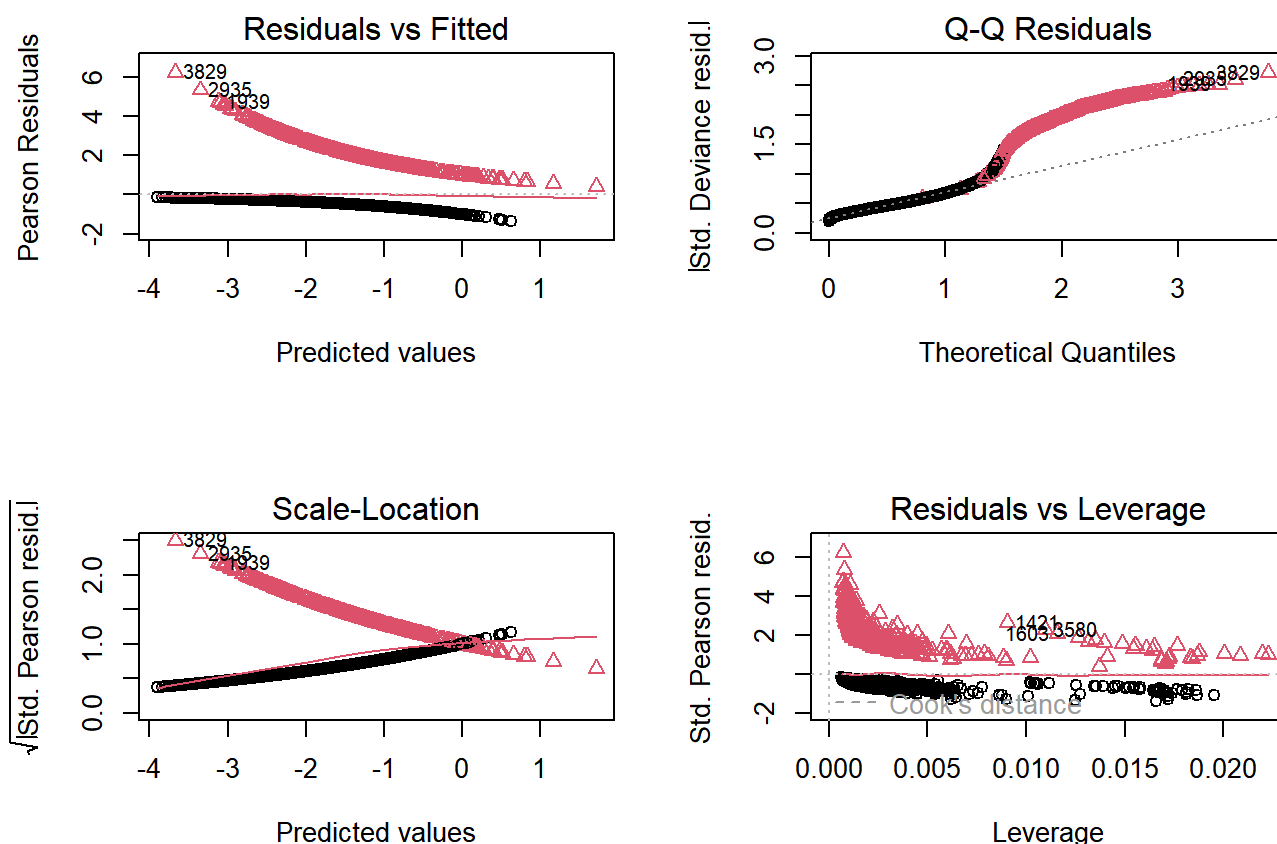
```
## 'log Lik.' 0.09380233 (df=6)
```

```
#Adjusted R2
n <- nrow(train) # number of observations
p <- length(coef(Model)) # number of coefficients
adj.R2 <- 1 - ((1 - pseudo.R2) * (n - 1) / (n - p - 1))
print(adj.R2)
```

```
## 'log Lik.' 0.09204386 (df=6)
```

The results of .094 for R^2 and .092 for R^2 adjusted, mean 9% of variability in the response variable CHD risk can be explained by our predictor variables. While this is relatively low, it is not entirely unexpected given the selection of predictor variables. Items were selected that aligned with the literature review, where other variables not selected can also affect CHD risk, such as total cholesterol and heart rate. Since R^2 is less commonly used in logistic regression, greater emphasis can be placed on evaluations like AIC and later AUC (Area Under the Curve) and still determine the model's predictive effectiveness. Nonetheless, R^2 remains a useful statistic to briefly discuss.

In logistic regression, residuals differ from linear regression since this model strives to predict probabilities versus actual values. This can be harder to interpret and calculate as logistic regression does not follow the same model assumptions like normal distribution that linear regression has.



The residual plots reveal two distinct clusters of points, primarily due to the binary nature of the response variable. Although these plots can be challenging to interpret at first, they offer valuable insights into the model. Red points represent individuals at-risk for CHD, while black points represent those not-at-risk. In the Residuals vs. Fitted plot, a noticeable curve in the residuals appears. In logistic regression, this is generally not a concern, as normality is not a required assumption. The bottom cluster, representing individuals not-at-risk, is centered around zero, suggesting that the model may more accurately predict those without CHD risk. Additionally, the horizontal zero line is relatively flat, indicating a good linear relationship between the variables. The Q-Q plot, which assesses normality, offers limited value in logistic regression, but again this plot highlights that extreme values correspond to cases with a higher likelihood of CHD risk. In the Scale-Location and Residual vs. Leverage plots, the same pattern persists. Points for at-risk individuals are farther from the horizontal axis. A slight curve in the Scale-Location plot may indicate heteroscedasticity, possibly due to outliers present. Outliers are present in all the plots and will be examined further. Lastly, the model's Residual Deviance, found in the model summary, is another measure of fit. It compares the likelihood of the current model with the null model, where lower deviance indicates a better fit. The Null Deviance is 2630.9, while the Residual Deviance is 2384.1, suggesting that the chosen predictors provide a good fit to the data.

Taking a look into the outliers shown on the plots. Outliers can be categorized as residuals with a value of less than -2 and greater than 2. When identifying outliers in logistic regression, deviance residuals are generally most useful because they reflect the contribution of each observation to the model's overall deviance.

```
residuals <- residuals(Model, type = "deviance")
Outliers <- which(abs(residuals) > 2) #singling out residuals less than -2 and greater than 2
Outliers <- as.data.frame(train[Outliers,])
all(Outliers$TenYearCHD == 1) #Confirming if all outliers found are participants at-risk of CHD
```

```
## [1] TRUE
```

```
nrow(Outliers)
```

```
## [1] 134
```

```
sum(train$TenYearCHD == 1)
```

```
## [1] 468
```

```
134/468
```

```
## [1] 0.2863248
```

To assess the outliers, the corresponding indices were put into a data frame to locate where in the dataset the extreme values are present. Upon reviewing the data frame, it became clear that many of these outliers were participants at-risk for CHD. A total of 134 outliers were identified, all of whom were at-risk subjects. The total count for those at-risk in our training data is 468, meaning 28.6% of the target group is being considered as outliers. These outliers will not be removed, as they represent a big portion of those at risk, and removing them could harm the model. This is a concerning find and requires more investigation.

Given that the residuals suggest the model struggles to predict CHD risk, along with the large number of outliers being at-risk participants, the model's predictive power should be further evaluated, to look for an underlying issue. This evaluation will involve testing the model on both its own data and the testing subset created earlier.

```
#Predicting with internal data
Mod.probs <- predict(Model, type = "response")
Mod.pred <- rep("0", 3099)
Mod.pred[Mod.probs > .5] = "1"
table(Mod.pred, train$TenYearCHD) #creating the confusion matrix
```

```
##
## Mod.pred    0    1
##           0 2612  443
##           1   19   25
```

```
mean(Mod.pred == train$TenYearCHD) #Calculating accuracy for internal data
```

```
## [1] 0.8509197
```

```
#Predicting with external data
Mod2.probs <- predict(Model, newdata = test, type = "response")
Mod2.pred <- ifelse(Mod2.probs > 0.5, "1", "0")
table(Mod2.pred, test$TenYearCHD) #creating the confusion matrix
```

```
##
## Mod2.pred    0    1
##           0 867 150
##           1   7  10
```

```
mean(Mod2.pred == test$TenYearCHD) #Calculating accuracy for external data
```

```
## [1] 0.8481625
```

The model predictions initially show high accuracy, at a first glance this would suggest strong performance. However, a closer examination of the confusion matrix reveals that most correct predictions are for participants not-at-risk, while predictions for at-risk are notably low. Upon further research into what could be causing this issue, it is apparent that the data is imbalanced. This makes sense as earlier it was shown that 18% of participants in the overall dataset were at-risk versus 82% not-at-risk. What is being seen in the model's prediction power is that there is not enough representation of those at-risk in the data to adequately train the model. This imbalance limits the model's ability to learn from the underrepresented group, and inflates the overall accuracy, giving a false impression of a well-performing model. This highlights the importance of checking for data imbalances early in the analysis process. Though it was shown the data had lower quantities of participants at-risk, it was under estimated how much it would affect the model. This is a valuable lesson I will carry forward with me for future analyses.

So how can this be fixed? There are a few methods to assist with imbalanced data including undersampling, oversampling, and weights. Undersampling reduces the majority class (not-at-risk), oversampling generates synthetic data to increase the minority class (at-risk), and weighting penalizes misclassifications of the minority class. All these methods were explored using the training dataset, among them the SMOTE (Synthetic Minority Oversampling Technique) method yielded the best results, improving predictions for at-risk participants while minimizing the impact on the model's explanatory power. However, all these methods came with trade-offs, including reduced goodness-of-fit and increased model complexity. The final model, enhanced with synthetic data, is presented below.

```
## package 'smotefamily' successfully unpacked and MD5 sums checked
##
## The downloaded binary packages are in
## C:\Users\knigh\AppData\Local\Temp\RtmpcRx9Ya\downloaded_packages
```

```
train$TenYearCHD <- as.factor(train$TenYearCHD)
set.seed(123)
smote_result <- SMOTE(X = train[, -16], target = train$TenYearCHD,
                      K = 5, dup_size = 3)
# Combine the SMOTE result into a new data frame
train_smote <- data.frame(smote_result$data)
names(train_smote)[ncol(train_smote)] <- "TenYearCHD"
train_smote$TenYearCHD <- as.factor(train_smote$TenYearCHD)

table(train_smote$TenYearCHD)
```

```
##
##      0      1
## 2631 1872
```

The dataset now includes 1,872 records representing participants at risk for CHD, helping to balance the discrepancy between at-risk and not-at-risk groups. To evaluate the predictive performance of the updated model, the testing data was used, as it offers a more reliable measure of the model's generalizability.

```
Model_S <- glm(TenYearCHD~gender+age+currentSmoker+diabetes+sysBP, data=train_smote, family=bino
mial)

Mod3.probs <- predict(Model_S, newdata = test, type = "response")
Mod3.pred <- ifelse(Mod3.probs > 0.5, "1", "0")
table(Mod3.pred, test$TenYearCHD)
```

```
##
## Mod3.pred    0    1
##           0 695  76
##           1 179  84
```

```
mean(Mod3.pred == test$TenYearCHD) # Accuracy 75.55%
```

```
## [1] 0.7533849
```

Although the model's accuracy dropped by 9.5%, its ability to correctly predict participants at risk for CHD has significantly improved. In the testing data, that has 160 at-risk participants, the previous model correctly identified only 10 cases, while the updated model correctly predicts 84. This is a 46.3% improvement for predicting CHD risk. However, there is still much more improvement needed, as the model's ability to identify those not-at-risk decreased by 25%. The model produced 179 false positives (Type I errors) and 76 false negatives (Type II errors). In this case, misclassifying at-risk participants (false negatives) is particularly more concerning, as it could result in missed opportunities for early intervention and prevention of serious heart disease. While the updated model shows promise, substantial improvements are still needed before using in a real medical setting. Re-sampling and collecting more data to better represent at-risk individuals would mostly likely offer the best method to enhance the model's performance.

Looking into the model summary and evaluation statistics we can get a further look into the new model's performance with the synthetic data.

```
summary(Model_S)
```

```
##
## Call:
## glm(formula = TenYearCHD ~ gender + age + currentSmoker + diabetes +
##       sysBP, family = binomial, data = train_smote)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -6.811457   0.287460 -23.695  < 2e-16 ***
## gender         0.499213   0.071065   7.025 2.15e-12 ***
## age           0.070888   0.004548  15.587  < 2e-16 ***
## currentSmoker 0.500475   0.072424   6.910 4.83e-12 ***
## diabetes      0.868042   0.197535   4.394 1.11e-05 ***
## sysBP         0.016940   0.001554  10.898  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 6113.9  on 4502  degrees of freedom
## Residual deviance: 5422.9  on 4497  degrees of freedom
## AIC: 5434.9
##
## Number of Fisher Scoring iterations: 4
```

The predictors in the updated model remain strong, with low p-values and standard errors, indicating their continued significance. The patterns of the coefficients stay consistent, showing positive relationships between the predictors and the response, while the negative intercept suggests CHD risk is unlikely without these predictors. The most notable change is the increase in the AIC value from 2396 to 5439, reflecting a loss of goodness-of-fit and increased model complexity. This increase may be attributed to the synthetic data altering the original relationships between variables, reducing the model's overall efficiency in explaining the dependent variable. While residuals could not be re-plotted due to the response variable being factored for SMOTE, the residual deviance in the model summary is still informative. Fortunately, the residual deviance is lower than the null deviance, indicating the model still performs well in predicting the response. Additional statistics McFadden R^2 , Adjusted R^2 , and AUC should also be evaluated.

```
## package 'pROC' successfully unpacked and MD5 sums checked
```

```
##
## The downloaded binary packages are in
## C:\Users\knigh\AppData\Local\Temp\RtmpcRx9Ya\downloaded_packages
```

```
##      Metric      Value
## 1 McFadden R² 0.1130207
## 2 Adjusted R² 0.1112995
## 3      AUC 0.7506436
```

The McFadden and adjusted R^2 have improved from .095 to .111, meaning that 1.6% more variability in our model can be explained. This is likely due to the synthetic data being derived from existing patterns in the original dataset. Another critical metric is the Area Under the Curve (AUC), which assesses the model's ability to

distinguish between classes (at-risk vs. not-at-risk). An AUC value ranges from 0 to 1, where 1 represents perfect classification and 0.5 suggests no better than random chance. In this model, an AUC of 0.75 indicates that there is a 75% likelihood the model correctly ranks an at-risk individual higher than a not-at-risk individual. This demonstrates a good, though not perfect, ability to differentiate between the two groups. While the model has lost some explanatory power, its predictive performance for identifying individuals at risk of CHD has improved overall.

In logistics regression, Confidence intervals (CIs) provide a range of values within which we expect the true prediction to lie, with a certain level of confidence. CIs for mean predicted and future predicted values at 95% confidence will be calculated. For the mean predicted value, we will be taking the average of all the predictors to simulate the average participant. For the future predicted value, a new combination of predictor values not present in the dataset will be used.

```
#Mean confidence intervals
# Average of all predictors
Av_data_point <- data.frame(gender = 1, age = 51, currentSmoker = 1, diabetes = 0, sysBP = 135)

# Predicting the mean predicted value and CI
prediction <- predict(Model_S, newdata = Av_data_point, type = "link", se.fit = TRUE)

lower_bound <- prediction$fit - 1.96 * prediction$se.fit
upper_bound <- prediction$fit + 1.96 * prediction$se.fit

# Display the result
c(lower_bound, upper_bound) # (-0.018, 0.199)
```

```
##           1           1
## -0.01816694  0.19911679
```

```
#Future confidence intervals
#New data point with different data
new_data_point <- data.frame(gender = 0, age = 70, currentSmoker = 0, diabetes = 0, sysBP = 145)

# Predicting the future predicted value and CI
prediction <- predict(Model_S, newdata = new_data_point, type = "link", se.fit = TRUE, interval = "confidence")

lower_bound_future <- prediction$fit - 1.96 * prediction$se.fit
upper_bound_future <- prediction$fit + 1.96 * prediction$se.fit

# Display the result
c(lower_bound_future, upper_bound_future) #(0.433, 0.781)
```

```
##           1           1
## 0.4334874 0.7806371
```

The CI for the mean predicted value, (-0.018, 0.199), indicates that the true probability of CHD risk for an average data point lies between -1.8% and 19.9%. Since this range includes 0, it suggests that the combination of mean predictor values does not strongly indicate CHD risk, further implying that average characteristics are not prominent enough to determine any risk. This aligns with our findings as more data points are for participants not-

at-risk. In contrast, the CI for the future predicted value, (0.433, 0.781), shows that the likelihood of CHD risk for this specific individual is between 43.3% and 78.1%. Though this interval is wider, it does not include 0, suggesting a stronger prediction. The increase in strength found in the future predicted value is likely the results from higher values like an age of 75. This highlights that the presence of CHD might be easier to uncover when more extreme values are present. These intervals are critical for evaluating the model's precision and reliability in making predictions.

To summarize the analysis, the initial dataset contained 4,133 participants, with 18% at risk of coronary heart disease (CHD). The low concentration of the targeted group created a class imbalance in the data. Despite this, the areas of interest including gender, age, smoking status, diabetes, Obesity and blood pressure still all showed a clear relationship with CHD risk. Male participants, those over 50, smokers, and individuals with high blood pressure or diabetes showed the highest risk. The predictor variables were then assessed for high correlation, retaining only those that aligned with the core areas of interest, aiming to simplify the model for better interpretability while preserving its relevance.

Before fitting the model, a training and testing subset was created for later evaluation. Upon fitting the initial model, it was found that BMI and diastolic pressure were not significant predictors. Multicollinearity was ruled out using the VIF function, and these variables were removed. The decision was based on the under-representation of overweight individuals and the weaker association of diastolic pressure with heart disease compared to systolic pressure. Removing these variables improved the model's overall fit. However, further analysis revealed that the model struggled with low R^2 , extreme residuals for at-risk participants, and poor accuracy in predicting at-risk individuals. This highlighted the impact of the imbalanced data. Further intervention was conducted to improve the model's ability to predict CHD risk more accurately.

To address the class imbalance, the SMOTE method was applied, increasing the number of participants at risk for CHD from 468 to 1,872, boosting predictions of at-risk individuals by 46.3%. However, this came with a 25% decrease in the precision of predicting not-at-risk individuals and a 9.5% drop in overall accuracy. With the increase in CHD risk representation, the amount of false negatives were greatly reduced, which in a medical setting is important to be able to address and prevent future heart complications. The model summary also showed that the predictors and relationships between variables overall remained strong, with the most notable change being the AIC increasing from 2396 to 5439, confirming the loss of goodness-of-fit and increased model complexity. However, other metrics like residual deviance, which remained lower than the null deviance, suggested the model still performed well. Additionally, R^2 improved by 1.6%, and the Area Under the Curve (AUC) was 0.75, indicating a decent ability to distinguish between the two risk categories. Lastly, The model's confidence intervals further highlighted that extreme values, such as older age and higher systolic pressure, improved the model's ability to predict CHD risk.

To conclude, the imbalance in the Framingham Heart Study dataset hindered the development of a strong regression model due to the under representation of individuals at risk for CHD. It is recommended to re-sample and include more of at-risk participants. Despite this, the dataset still provided valuable insights into the relationships between key predictors and CHD risk. Our model confirmed the expected positive associations between age, gender, smoking status, diabetes, and systolic blood pressure with increased CHD risk. This can be found most evident in the relationship graphs towards the beginning of the analysis that shows which groups are at bigger risk of the disease. As you increase in age, the body is more susceptible to disease. Men tend to experience more heart disease complications than women. Additionally, increases in body weight, particularly due to fatty deposits, and elevated blood pressure significantly contribute to the risk of heart disease. Furthermore, coronary heart disease is the leading cause of death among individuals with diabetes. These findings align with the literature review, underscoring the importance of these variables in predicting heart disease. Revisiting odds ratios for the enhanced model can quantify the strength of these relationships.

```
exp(coef(Model_5))
```

```
##      (Intercept)          gender          age currentSmoker      diabetes
## 0.001101088    1.647424712    1.073460944    1.649504323    2.382242152
##           sysBP
## 1.017084730
```

The odds for men to have CHD risk is 1.647 times higher or 64.74% more than women. Current smokers have 64.95% higher odds of CHD risk compared to non-smokers, and individuals with diabetes have a 138% increase in odds. For continuous variables, systolic blood pressure (sysBP) has an odds ratio of 1.017, indicating a 1.7% increase in CHD odds for each 1 mmHg rise, while age increases 7.3% in odds per additional year. These odds show a slight increase compared to our initial model. These results show a clearer understanding of the predictors and their relationship to CHD risk, especially with more representation of the at-risk group.

While the imbalance in the dataset limited the development of a robust regression model, valuable insights into the relationships of the key predictors of CHD risk were still obtained. By addressing the imbalance in the data and incorporating more at-risk participants, future models may provide even clearer insights into these relationships and improve predictive accuracy overall.

In this report, well over 20 hypothesis tests were conducted to fit and improve the regression model. As the exact number of hypothesis tests cannot be clearly determined, the number 20 will be denoted as m . If m was multiplied by every p -value in this report, my analysis of the predictor variables would not change. The p -values of the final predictor variables are very low, even if multiplied by 20, would still remain under the .05 threshold. However, for variables BMI and DiaBP removed in the earlier models, it would become even more apparent of their insignificance as the p -values would be well over the value of 1.